Calibration Technician Training Module 006

Data Quality, Refusal, Sensor Issues, and Troubleshooting Boundaries

Field	Value
Document type	Workspace 006-B technician training module
Module	006 - Data Quality, Refusal, Sensor Issues, and Troubleshooting Boundaries
Version	v0.1 Review Deliverable
Prepared	July 2026
Training lane	Calibration Technician Training - not customer prospecting
Status	Internal training draft; not a final field SOP, engineering interpretation guide, or analyzer manufacturer manual

Locked module purpose
Teach a calibration technician how to recognize questionable data, protect the instrumented rod and sensors, respond to refusal or rapid advancement, and know when to correct, rerun, document, or escalate. Module 004 taught software workflow. Module 005 taught field measurement accuracy. Module 006 teaches what to do when the data or field condition is not clean.

Core rule
A calibration technician does not force questionable data into the report path just to finish faster.

1. Source basis and training boundary

This module is based on the following active training sources and SME corrections:

- Chapter 2 Draft and Field Lessons.
- Calibration Data Collection Cheat Sheet.
- SME clarification on sensor pulse check, refusal, rapid advancement, instrumented rod protection, and field correction.
- Calibration Service Module Extraction Worksheet.
- Module 004 - Field Setup and SPT Analyzer Software Workflow.
- Module 005 - LP, LE, Depth Intervals, and Field Measurement Accuracy.

Chapter 2 provides the field lesson base: loose sensors, shallow bedrock, refusal, hammer RPM, hydraulic pressure, hammer cycling, full 30-inch stroke, clean/tight/flush rod joints, rod length measurement, and tech control of data quality. The cheat sheet provides the software path for data collection, end-test behavior, and repeated intervals with updated LP and LE.

Training boundary
This module does not make the new tech responsible for engineering interpretation. The technician is not deciding final hammer pass/fail judgment, foundation design implications, soil classification, liquefaction analysis, final N60 interpretation, or whether a customer project data set is acceptable for engineering use.

The technician is responsible for recognizing questionable data, protecting equipment, correcting obvious setup issues, rerunning when the system checks out, documenting what happened, and escalating when the issue exceeds field authority.

2. Learning outcomes

1. Explain what makes a data set questionable.
2. Recognize loose sensor, failed pulse check, or poor sensor-response indicators.
3. Explain the sequence for troubleshooting a failed pulse check.
4. Recognize refusal and understand why refusal data is not normal calibration data.
5. Recognize rapid advancement and protect the instrumented rod.
6. Check common field causes of bad data before blaming the hammer.

7. Understand why rod joints must be clean, tight, and flush.
8. Recognize hammer-operation concerns such as abnormal idle/RPM behavior, hydraulic leaks, cycling issues, or short stroke.
9. Decide when to proceed, rerun, document, or escalate.
10. Explain why the tech controls data quality but does not override project sampling requirements.

3. What good data means

Good data is not just a saved file. Good data means the field setup, software setup, handwritten record, and collection conditions all support the same story.

- Correct customer / rig / borehole information.
- Correct depth interval and sample interval.
- Correct LP and measured LE.
- Correct rod type.
- Sensor connection verified in software.
- Pulse check passed before live collection.
- Rod and anvil connections snug / flush.
- Hammer operating normally for that rig and operator.
- Blow counts recorded every 6 inches.
- Refusal, rapid advancement, or abnormal field conditions documented.

Training rule
Good data can be reviewed by someone who was not standing at the rig.

4. Questionable data

A data set may be questionable if any of the following occur:

- Pulse check does not produce the expected graph shape.
- One or more sensors is loose, disconnected, dirty, damaged, or not tightened correctly.
- Depth interval is wrong.
- LP is copied from a prior run without confirmation.
- LE is guessed.
- Blow counts are missing.
- Refusal occurs.
- Rapid advancement occurs.
- The instrumented rod nearly enters the hollow stem auger or borehole.
- Rod joints are loose, dirty, or not flush.
- Anvil is not threaded flush into the instrumented rod.
- Hammer is releasing early or cycling inconsistently.
- Idle/RPM or hydraulic behavior appears outside normal operation for that rig.
- The customer/driller rushes the sequence and setup is compromised.

Training rule
Questionable does not always mean useless, but it must be documented and reviewed.

5. First response to questionable data

When data looks off, the tech should not immediately tell the customer the hammer is bad. Correct first response:

11. Stop and check setup.
12. Confirm sensor connection.
13. Confirm accelerometers are secure.
14. Repeat the pulse check if needed.
15. Confirm rod joints are clean, tight, and flush.
16. Confirm anvil is threaded flush into the instrumented rod.

17. Confirm LP, LE, and depth interval.
18. Confirm hammer is operating normally with the operator.
19. Rerun or collect another interval if the issue is corrected and the pulse check passes.
20. Document what happened.
21. Escalate if the issue remains unexplained or equipment damage is suspected.

Training rule
Bad-looking data is a stop-and-check signal, not an instant hammer diagnosis.

6. Pulse check problems

The pulse check confirms sensor response before live data collection. The icon may look like a top hat because it visually represents the shape the graph should show, but the training term is pulse check.

When the pulse check is pressed, the analyzer sends an electrical pulse through the sensors and back to the software.

Correct result
The graph shows the expected top-hat shape. This means the sensor path appears connected and ready.

Problem result
If the graph does not show the expected shape, one or more sensors may be disconnected, loose, not tightened correctly, affected by a dirty connector, affected by a cable/connector issue, or damaged.

Failed pulse check troubleshooting order	Required action
1	Reconnect/check cables.
2	Check sensor and pigtail connections.
3	Restart the software completely.
4	Replace accelerometers/sensors with backups if needed.
5	Rerun the pulse check.
Good pulse check after troubleshooting	Proceed with data collection if the physical setup is also ready.
Backup sensors also fail	Likely cord or connector issue. Stop and escalate to the office.

Do not proceed after failed pulse check
Do not press pause and start live data collection until the pulse check looks correct.

7. Sensor and accelerometer issues

Loose sensors are a leading cause of poor data. Sensors should be secured firmly, but not overtightened.

- Accelerometers are secure.
- Accelerometer tightening hardware is used correctly.
- Sensor cable is connected.
- Pigtail connection is secure.
- Connector is not visibly dirty or damaged.
- Sensor boxes/checkmarks are present in the Sensor tab.
- Pulse check graph looks correct.

Training rule
Tight enough is not the same as overtightened.

Damage type	Field response
Accelerometer damage	Inspect for cable or connector damage. If no cable damage is discovered and backup accelerometers are available, replace the damaged accelerometer, repeat the pulse check, and continue only if the setup returns to valid operation. Call the office and document the damage.

Suspected strain gauge damage	Stop and escalate immediately. Strain gauges are built into the instrumented rod body and are not field-replaceable. Call the office immediately and communicate with the drill manager or senior authority path before continuing.

8. Rod and anvil connection issues

Rod joints matter because energy moves through the rod system. Rod joints must be clean, tight, and flush; small gaps reduce energy transfer.

- Rod joints clean.
- Rod joints tight.
- Rod joints flush.
- Correct rod type.
- No obvious damaged rod sections.
- Instrumented rod snug to rod string.
- Anvil threaded flush into top of instrumented rod.

Training rule
A small connection problem can become a data problem.

9. Hammer operation issues

The technician is not rebuilding or fully diagnosing the hammer in this module, but must recognize when hammer operation may be affecting data.

- Hammer not cycling consistently.
- Hammer appears to short stroke.
- Hammer not releasing cleanly.
- Idle/RPM concern for that rig.
- Visible hydraulic leak.
- Hydraulic gauge/pressure concern if escalated in real time.
- Hammer behavior changes during collection.

RPM / idle training rule
There is no universal hammer RPM or idle number for all rigs. The operator usually knows the idle point normally used for hammer operation. If a problem persists and hammer operation seems involved, try a different idle point with the operator before assuming hammer failure.

Hydraulic pressure training rule
The tech mainly checks for obvious hydraulic issues such as leaks or abnormal behavior. If the issue escalates to a phone call, hydraulic pressure/gauge information should be verbally given in real time if available.

Short stroke definition
Short stroke means the hammer is releasing before reaching the designated 30-inch mark.

Customer language boundary
Do not tell the customer the hammer has failed unless that conclusion has been confirmed through the proper authority path. Safer wording: "We are seeing something that needs to be checked before we treat this as clean data."

10. Refusal

Refusal is a concrete field condition.

Refusal rule
Refusal means the sampler does not advance a full 6 inches in 50 blows.

Refusal data should not be treated as normal clean calibration data.

- Record the blow count.
- Record how far the sampler advanced.
- Record the depth interval.
- Document the field condition.
- Do not treat the interval as normal clean calibration data.
- Follow the proper field decision / escalation path.

Training rule
Refusal is not just a drilling delay. It changes the quality and usefulness of the calibration data.

11. Shallow bedrock or obstruction

Shallow bedrock or obstruction may require shallower testing, shorter intervals, or nearby boreholes if needed to collect enough representative data.

- Depth where issue occurred.
- Blow count behavior.
- Whether sampler advanced.
- Whether refusal occurred.
- Whether drilling moved to another borehole.
- Whether interval was shortened.
- Whether additional data was collected elsewhere.

Training rule
The solution is not to force bad data. The solution is to document the condition and collect usable data where possible.

12. Rapid advancement / loose soil

Not every data-quality problem is hard ground. Loose soil can cause the rod and sampler to advance very quickly.

CRITICAL WARNING - protect the instrumented rod
Do not let the instrumented rod enter the hollow stem auger or borehole. This is one of the fastest ways to damage sensors.

The tech should observe the first 6 blows closely to confirm the soil is not so loose that the rod and sampler are advancing too quickly.

Before collection begins, the tech must clearly tell the operator:

Operator stop instruction
Stop hammer operation immediately if the instrumented rod is in danger of entering the hollow stem auger or borehole.

Training rule
Protecting the instrumented rod is part of data quality.

13. Production-job pressure

On production jobs, calibration must work within the project sampling requirements. Calibration can control data quality, but it does not override the standards set by the entity paying for the drilling.

- Project sample intervals may control the interval used.
- Project sampling frequency may control when samples are taken.
- The calibration tech may prefer 24-inch intervals but may need to collect 18-inch intervals.
- If insufficient calibration data is collected during production sampling, additional standard collection may be done later if allowed.

Training rule
Do not sacrifice data quality for production speed, but do not ignore project sampling requirements either.

14. Proceed, rerun, document, or escalate

Action	Use when
Proceed	The setup issue has been corrected, the pulse check now looks good, the physical setup is ready, and there is no suspected equipment damage or unresolved data-quality issue.
Rerun / collect another interval	A correctable setup, interval, sensor, or documentation issue was found and corrected, and another valid data set can still be collected. If an interval mistake is caught on site, best practice is to correct it in the field by drilling deeper and collecting another data set with proper interval, LP, LE, and 6-inch blow-count documentation.
Document	Refusal, interval changes, move to another borehole, 18-inch interval instead of preferred 24-inch interval, shallow bedrock, obstruction, questionable saved data, customer/driller pressure, or a field mistake discovered after collection.
Escalate	Data remains questionable after setup checks; pulse check fails after cable/software/back-up sensor troubleshooting; strain gauge damage is suspected; instrumented rod/analyzer kit damage occurs; hammer behavior remains abnormal; mechanical failure occurs; or tech is unsure whether continuing will produce valid data.

Pulse check decision rule
Correctable issue + good pulse check = proceed. Backups fail ping too = likely cord or connector issue = escalate.

15. Standard tech authority boundary

A standard calibration tech does not independently decide whether to continue, abort, or reschedule after major equipment damage or unresolved data problems.

- Call the office / senior calibration tech.
- Explain the severity.
- Document what happened.
- Wait for direction when the issue exceeds field authority.

A senior calibration tech may make immediate field judgment when necessary, but the issue still must be communicated to the office.

Training rule
When in doubt, preserve the data, protect the equipment, and escalate.

16. Field notes that support troubleshooting

Useful notes include:

- Pulse check failed / passed after correction.
- Sensor was loose.
- Accelerometer was tightened or replaced.
- Connector was dirty.
- Rod joint was not flush.

- Anvil was not flush.
- Hammer cycling was inconsistent.
- Suspected short stroke.
- Idle/RPM concern or alternate idle point tried.
- Hydraulic leak, gauge, or pressure information during escalation call.
- Refusal.
- Rapid advancement.
- Shallow bedrock.
- Obstruction.
- Moved to another borehole.
- Project required 18-inch interval.
- Collected extra standard interval after production sampling.
- Customer/driller pressure.
- Test rerun.
- Questionable data saved for review.

Training rule
The report builder can only explain what the technician records.

17. Common mistakes

Mistake	Why it matters	Correct behavior
Pressing pause after failed pulse check	Sensors may not be reading properly	Correct issue and repeat pulse check before collection
Calling hammer failure too early	Setup issues often cause bad data	Check sensor, rod, anvil, LP, LE, depth, and hammer operation first
Using refusal data as normal data	Refusal changes data quality	Document and follow field decision path
Ignoring rapid advancement	Instrumented rod/sensors can be damaged	Watch first 6 blows and stop hammer if needed
Overtightening sensors	Can cause damage or bad setup	Secure firmly; do not overtighten
Ignoring rod joint gaps	Gaps reduce energy transfer	Clean, tighten, and flush connections
Not documenting project interval requirements	Mixed intervals may look like mistakes	Note project-driven intervals
Deciding alone after damage	Standard tech lacks authority	Call office / senior tech

18. Training scenarios

Scenario A - rapid advancement

A tech begins a calibration run at 20 ft. The pulse check looks correct. The operator starts hammering, but the sampler advances very quickly during the first few blows. The instrumented rod is getting close to the hollow stem auger. The operator is focused on production speed and keeps going.

Correct trainee response should include:

- Stop hammer operation before the instrumented rod enters the hollow stem auger or borehole.
- Protect the instrumented rod and sensors.
- Document rapid advancement and depth interval.
- Do not treat the run as clean data if rod/sensor risk affected collection.
- Correct the setup or choose another interval if possible.
- Communicate clearly with the operator before restarting.
- Escalate if equipment may have been damaged or if there is uncertainty about continuing.

Scenario B - failed pulse check

The tech reaches the data collection screen, but the pulse check does not produce the expected graph shape. The tech reconnects cables, checks the sensor/pigtail connections, restarts the software, replaces accelerometers with backups, and reruns the pulse check. The backups still do not produce a good ping.

Correct trainee response should include:

- Do not begin live data collection.
- Recognize that the failure after backup replacement is likely a cord or connector issue.
- Stop and escalate to the office.
- Document the troubleshooting steps already completed.
- Communicate any visible connector, cable, or kit damage.

19. Knowledge check

#	Question	Answer key
1	What should a tech do when data looks bad at the first depth?	Stop and check setup before blaming the hammer.
2	What does the pulse check verify?	That the sensor signal path is responding correctly and producing the expected graph shape.
3	Should live data collection begin after a failed pulse check?	No. Correct the issue and repeat the check first.
4	What should happen if cables are reconnected, software is restarted, backup sensors are installed, and the pulse check still fails?	Stop and escalate to the office because the issue is likely a cord or connector problem.
5	Why are loose sensors a problem?	Loose sensors are a leading cause of poor data.
6	Should sensors be tightened as hard as possible?	No. Secure firmly, but do not overtighten.
7	What is short stroke?	The hammer releases before reaching the designated 30-inch mark.
8	Is there one universal hammer RPM/idle value for all rigs?	No. The operator usually knows the normal idle point for hammer operation on that rig.
9	What is the refusal rule?	Sampler does not advance a full 6 inches in 50 blows.
10	Should refusal data be treated as normal calibration data?	No. It must be documented and handled as a data-quality issue.
11	What should the tech watch during the first 6 blows?	Whether the rod/sampler is advancing too quickly and risking the instrumented rod entering the HSA or borehole.
12	Why do rod joints need to be clean, tight, and flush?	Small gaps reduce energy transfer and can affect data quality.
13	What should a tech do if accelerometer damage occurs but backups are available and cables are not damaged?	Replace with backups, repeat pulse check, continue only if valid operation returns, call office, and document the damage.
14	What should happen if strain gauge damage is suspected?	Stop and call the office immediately; communicate with the drill manager or senior authority path before continuing.
15	What is the safest rule when unsure?	Preserve the data, protect the equipment, and escalate.

20. Practical signoff checklist

Signoff item	Pass standard	Trainer initials/date
Recognizes questionable data	Trainee identifies setup, sensor, rod, hammer, refusal, and rapid advancement causes	
Handles failed pulse check	Trainee follows cable/software/backup sensor sequence and escalates if backups still fail	
Checks sensor setup	Trainee secures sensors firmly without overtightening	
Handles accelerometer damage	Trainee knows when to replace with backups, retest, continue, call office, and document	
Handles strain gauge concern	Trainee stops and escalates immediately because strain gauges are not field-replaceable	
Checks rod/anvil setup	Trainee confirms clean/tight/flush rod joints and flush anvil connection	
Identifies refusal	Trainee states refusal rule and explains why refusal data is not normal data	
Protects instrumented rod	Trainee watches first 6 blows and communicates stop condition to operator	
Recognizes hammer-operation concerns	Trainee identifies idle/RPM, hydraulic leak/pressure, cycling, and short-stroke	

	concerns	
Documents field issues	Trainee records notes that a report builder can use	
Knows authority boundary	Trainee escalates damage or unresolved questionable data instead of deciding alone	

21. SME-needed / next module inputs

Module 006 is complete enough for a v0.1 training deliverable. No SME-needed item is currently blocking this module.

Items that should feed later modules:

- More detailed automatic hammer mechanical explanation should remain tied to KC014 Automatic Hammer Systems.
- Final report package/reviewer responsibilities should remain in a later report handoff module.
- Formal trainer approval requirements and minimum observed/supervised calibration counts should remain in a later training progression module.

Revision history

Version	Status	Change
v0.1	Review deliverable	Created from Module 006 thread draft and SME corrections on failed pulse checks, RPM/idle handling, hydraulic-pressure escalation, short-stroke wording, proceed/rerun authority, accelerometer damage, and strain gauge escalation.